



# NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

## Second Semester M. Tech (MMD) (CBCS) Degree Examinations

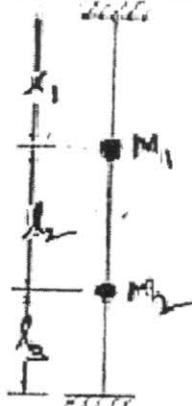
Summer Semester July 2024

### 22MMD201 – THEORY OF VIBRATIONS

Duration: 3 Hours

Max. Marks: 100

**Note:** Answer **Five full** questions choosing **One full** question from **each Unit**.

| Unit – I                                                                           |                                                                                                                                                                                                                                                                                                          | Marks | BT* | CO* | PO* |
|------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|-----|-----|
| 1.                                                                                 | a) Add the following harmonic motions analytically and check the solution graphically - $x_1 = 3 \sin(\omega t + 30^\circ)$ and $x_2 = 4 \cos(\omega t + 10^\circ)$                                                                                                                                      | 10    | L*3 | 1   | 1   |
|                                                                                    | b) In a spring-mass system, the mass of 10 kg makes 40 oscillations in 20 seconds, without damper. With damper, the amplitude decreases to 0.2 of the original value after 5 oscillations. Find out: i) Stiffness of the spring, ii) logarithmic decrement, iii) damping ratio and iv) damping constant. | 8     | L3  | 1   | 1   |
|                                                                                    | c) What is Displacement transmissibility?                                                                                                                                                                                                                                                                | 2     | L1  | 1   | 1   |
| 2.                                                                                 | a) Define Transmissibility ratio? Derive an expression for the same. Also derive the expression for the angle through which the transmitted force lags the impressed force.                                                                                                                              | 8     | L2  | 1   | 1   |
|                                                                                    | b) Derive the equation of motion, natural frequencies and mode shapes for the masses fixed on a tightly stretched string as shown in figure. 2b.                                                                                                                                                         |       |     |     |     |
|  |                                                                                                                                                                                                                                                                                                          | 12    | L3  | 1   | 1   |
| Unit – II                                                                          |                                                                                                                                                                                                                                                                                                          |       |     |     |     |
| 3.                                                                                 | a) What is Vibration isolation? How do you achieve the same? Derive an expression for transmissibility ratio of a vibration isolation system with rigid foundation.                                                                                                                                      | 10    | L2  | 2   | 1   |
|                                                                                    | b) Find the response of a compacting machine, modeled as a SDOF system, when subjected to a sudden application of the pressure modeled as a step force.                                                                                                                                                  | 10    | L3  | 2   | 1   |
| 4.                                                                                 | a) An air compressor of mass 200 kg, with an unbalance of 0.01 kg-m is found to have a large amplitude while running at 1200 rpm. Determine the mass and spring constant of the absorber to be added if the natural frequencies of the system are to be at least 20 % from the impressed frequency.      | 8     | L3  | 2   | 1   |
|                                                                                    | b) With a neat sketch explain the working of a Lanchester damper.                                                                                                                                                                                                                                        | 7     | L2  | 2   | 1   |
|                                                                                    | c) What is the significance of Convolution Integral.                                                                                                                                                                                                                                                     | 5     | L2  | 2   | 1   |

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## Unit – III

- |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |    |    |   |   |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|---|---|
| 5. a) | Compare different types of transducers used in vibration measurement.                                                                                                                                                                                                                                                                                                                                                                                                    | 7  | L2 | 3 | 1 |
| b)    | With some examples, discuss the nature of non-linearity in some physical systems.                                                                                                                                                                                                                                                                                                                                                                                        | 7  | L2 | 3 | 1 |
| c)    | A device used to measure torsional acceleration consists of a ring having a moment of inertia of $0.049 \text{ kg-m}^2$ connected to a shaft by a spiral spring having a scale of $0.98 \text{ N-m/rad}$ and a viscous damper having a constant of $0.11 \text{ N-m-s/rad}$ . When the shaft vibrates with a frequency of $15 \text{ cpm}$ , the relative amplitude between the ring and shaft is found to be $2^\circ$ . What is the angular displacement of the shaft? | 6  | L3 | 3 | 2 |
| 6. a) | Discuss Lindstedt's Perturbation method to find the solution of the pendulum equation, considering it to be a periodic solution.                                                                                                                                                                                                                                                                                                                                         | 10 | L2 | 3 | 1 |
| b)    | A commercial vibration pickup has a damped natural frequency of $4.5 \text{ Hz}$ and a damping ratio of $0.5$ . What is the range of impressed frequency at which the amplitude can be read directly from the pickup with an error not exceeding $2\%$ of the actual amplitude?                                                                                                                                                                                          | 6  | L3 | 3 | 1 |
| c)    | Compare different frequency measuring instruments.                                                                                                                                                                                                                                                                                                                                                                                                                       | 4  | L2 | 3 | 1 |

## Unit – IV

- |       |                                                                                                                                                                                                                                                                                                      |    |    |   |   |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|---|---|
| 7. a) | What is Power spectral density of a stationary random process? Discuss its properties. How do you differentiate between wide band and narrow band processes?                                                                                                                                         | 10 | L2 | 4 | 1 |
| b)    | Discuss and differentiate between mechanical and electromagnetic exciters.                                                                                                                                                                                                                           | 10 | L2 | 4 | 1 |
| 8. a) | The spectral density of a random signal is given by $S(f) = 0.0001 \text{ m}^2/\text{cycle/s}$ for $f$ between $10 \text{ Hz}$ and $1000 \text{ Hz}$ and $0$ elsewhere. Find the standard deviation and the root mean square value of the signal by assuming its mean value to be $0.05 \text{ m}$ . | 5  | L3 | 4 | 1 |
| b)    | Discuss the use of digital signal processing in analysis of random signals.                                                                                                                                                                                                                          | 5  | L2 | 4 | 1 |
| c)    | If $x$ and $y$ are statistically independent then $E[xy] = E[x] E[y]$ . If $z = x + y$ , where $x$ and $y$ are statistically independent, show that $E[z^2] = E[x^2] + E[y^2] = 2E[x][y]$ .                                                                                                          | 6  | L3 | 4 | 1 |
| d)    | Develop the Parseval's formula for periodic functions.                                                                                                                                                                                                                                               | 4  | L2 | 4 | 1 |

## Unit – V

- |        |                                                                                                                                       |    |    |   |   |
|--------|---------------------------------------------------------------------------------------------------------------------------------------|----|----|---|---|
| 9. a)  | Derive the frequency equation for free vibrations of a string with both ends fixed.                                                   | 10 | L3 | 5 | 1 |
| b)     | Why is an unrestrained system called a semidefinite system?                                                                           | 10 | L2 | 5 | 1 |
| 10. a) | Determine the equation of motion for vibration of membranes.                                                                          | 5  | L2 | 5 | 1 |
| b)     | Determine the natural frequencies of the triple pendulum shown in figure. 10b. Take $l_1 = l_2 = l_3 = l$ and $m_1 = m_2 = m_3 = m$ . |    |    |   |   |

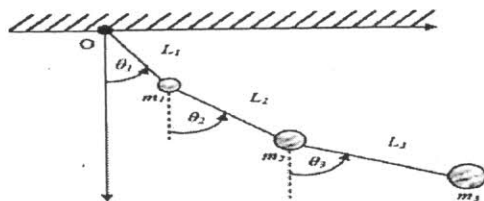


Figure 10(b).

|    |    |   |   |
|----|----|---|---|
| 15 | L3 | 5 | 1 |
|----|----|---|---|